The radical counts each active prime once

8 — 2 · 2 · 2 — power = 2^3, radical = 2

9 — 3 · 3 — power = 3^2, radical = 3

$72 = 2^3 \cdot 3^2 \quad \rightarrow \quad \text{rad}(72) = 2 \cdot 3 = 6$